

Clustering

Classification without labels

In unsupervised learning, we are given data without labels

- We want **to find groups of similar objects** → **clustering**
- **Many clustering algorithms** exist

Issues for clustering

The **major issues** in clustering are:

- **Representation**: **how** we **represent** examples
- **Clustering type**: **flat** or **hierarchical** structure
- **Similarity metric**: **how to measure** whether an example is **part of a cluster**
- **Number of clusters**: **how to define** the **number** of clusters

Classes of clustering algorithms

Two **approaches** to distinguish clustering algorithms

- **Flat** vs **Hierarchical algorithms**
 - **Flat**: clusters **don't intersect**
 - **Hierarchical**: **tree structure**, **shows** the **relations** between clusters
- **Hard** vs **Soft clustering**
 - **Hard**: **each example** belongs to exactly **one cluster**
 - **Soft**: **an example** can belong to **one or more clusters**

K-means clustering

A **flat algorithm** based on **hard clustering**

General idea

Given k clusters, **we aim to find a partition of data into k sets**
minimizing the **variation within each set**, defined as follows:

$$\min_{C_1, \dots, C_k} \sum_{j=1}^k V(C_j)$$

Variation is typically given by: $V(C_j) = \sum_{i \in C_j} \|x_i - \mu_j\|^2$

Centroids are computed with: $\mu_j = \frac{1}{|C_j|} \sum_{i \in C_j} x_i$

💡 K-means **assumes** the **same variance in every direction**

- Assumes **spherical clusters**
- **Non-spherical clusters** are **impossible** to identify

Algorithm

The algorithm follows these **steps**:

- (Input) Data points $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$
- We **fix** a **number of clusters** k
- We **get** some **initial cluster centroids** (= centers) $\mu_1, \dots, \mu_k$
- **While clusters change**:
 - We **assign each datapoint** to the **closest centroid** forming new clusters

$$C_j = \{i \in \{1, \dots, n\} : j = \arg \min_l \|x_i - \mu_l\|\}$$

- **Compute new clusters centroids**

Properties

Sensitive to the scale of features

- Feature **normalization** is **required**

Is **guaranteed to converge**:

- It **strictly improves** the **objective**
(if there is at least one cluster change and the set of possible partitions is finite)

Is **guaranteed to find a local minimum** (NOT a global one)

- The function is generally not convex and usually has many minima

Initial centroid selection

Common **heuristics to define initial centroids** are

- **Random points** in the space
- **Randomly picked examples**
- **Points least similar to any existing center**
- **Multiple starting points**
- **Initialized** with the **results of another clustering method**

Running time

Given k clusters and n datapoints:

- **Assignment:** $O(kn)$
- **Centroid computation:** $O(n)$

Distance issue

The metric used **to measure distance** depends on the goal and the datapoints

- **Euclidian distance:** is the absolute distance between two vectors
- **Cosine similarity:** expresses the similarity between two vectors (= measures the angle)
- **Cosine distance**

EM clustering

A **flat algorithm** with **soft clustering**

General idea

Very similar to K-means but:

- **Assumes elliptical clusters** → **expects data** to come **from** a mixture of **Gaussians**
 - **Centroids get a weighted contribution** from points
 - **Different variance directions** have **different weights** on the result
- Is a **soft clustering algorithm**
 - **Assigns** to each datapoint **the probability to be part of each cluster**

The model is parameterized as the **probability of the point x_n to be part of the model**:

$$p(x_n) = \sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k) \quad \text{subject to} \quad \sum_{k=1}^K \pi_k = 1, \pi_k \geq 0$$

Where:

- π_k are mixing **coefficients**, represent the **weight of the cluster**
- $\mathcal{N}(\dots)$ **indicates a Gaussian distribution**
 - The **model** is a **linear combination of k Gaussian distributions**
- μ_k is the **mean of the Gaussian distribution**
 - Indicates **the center of the cluster**
 - Indicates the **peak of the Gaussian distribution**
- Σ_k is the **covariance**
 - **Defines the shape and the direction** of the cluster
 - **Allows** the existence of **elliptical clusters**

Spectral clustering

General idea

Allows to cope with **non-spherical geometries**

It's **based on** the **construction of a graph**

- **Each datapoint** is a **node**
- **Each node** is **connected to its** k **nearest neighbors by edges**
 - **To measure similarity** it is common to use a **Gaussian Kernel**

$$W(i, j) = \exp\left(\frac{-|x_i - x_j|^2}{\sigma^2}\right)$$

Then **3 fundamental matrices** are **computed**:

- **Adiacence matrix** A : indicates the **connections between nodes**
- **Degree matrix** D : indicates the **number of connections per node**
- **Laplace matrix** $L = D - A$: contains **all information about the graph's structure**

From the Laplace matrix L **eigenvectors** are **computed**

- Equals to dimensionality reduction
- The **number of eigenvector with eigenvalue** equal to 0 indicates
 - The **number of connected components** in the graph
 - The **number of clusters** in the dataset
- A **new representation** of the dataset is obtained

A simple clustering algorithms can be applied to the new representation of data

Hierarchical clustering

Produces a **set of nested clusters**

- **Organized** as a **hierarchical tree** → called **dendrogram**
- 2 approaches:
 - **Bottom-up / agglomerative**
 - **Top-down / divisive**

We will focus on the **agglomerative approach**

General idea

- **Merges** very **similar instances**
- **Incrementally builds larger clusters** out of smaller ones

Algorithm

The algorithm follows these **steps**:

- Initially **each instance** is **its own cluster**
- Repeat:
 - **Pick** the **2 closest clusters**
 - **Merge** them in a **new cluster**
 - **Stop** when there is **only one cluster left**

There are **multiple ways to define “closest”** for clusters with multiple elements
(closest pair, farthest pair, average among all distances)

Summary

	K-Means	EM	Spectral	Hierarchical (Agglomerative)
Assignment	Hard	Soft (probabilistic)	Hard (via K-Means)	Hard
Cluster shape	Spherical	Ellptical	Any	Any (linkage-dependent)
Requires k	Yes	Yes	Yes	No
Scalability	Excellent	Moderate	Poor	Poor
Key strength	Fast, scalable	Soft uncertain membership	Non-convex clusters	No k needed, hierarchy
Typical use	Large-scale data	Overlapping clusters	Non-linear-shapes	Gene expression